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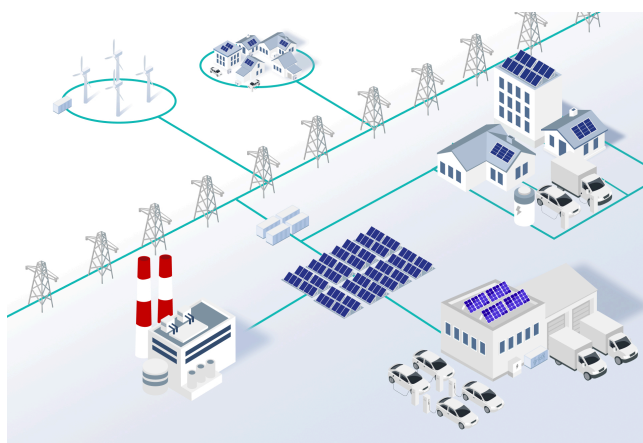
Energy Flexibility

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Energy flexibility refers to the ability to adjust power generation and/or demand. Power flows must be adjusted to account for varying electricity prices, electricity market changes and/or transmission and distribution system conditions and regulation. Energy flexibility is crucial to enable the transition of our energy system to renewable energy sources. Flexibility can be provided by both large assets and small-scale residential assets.

The difference between implicit and explicit flexibilities



A fully renewable energy system needs all kinds of DERs to provide flexibility to stabilize the grid

Energy flexibility manifests in two primary forms – implicit and explicit. Implicit flexibility is the consumer's response to price signals, – either manually or via a smart energy management system – empowering **prosumers** to optimize their energy usage and to save costs, for example by using **flexible energy tariffs**.

Explicit flexibility, on the other hand, is a committed, dispatchable resource that can be traded on various energy markets, such as wholesale, balancing, system support, and reserves markets. This type of flexibility is vital for balancing and managing grid congestion and is used by Balance Responsible Parties (BRPs) and Distribution System Operators (DSOs) to reduce their costs.

Why value stacking becomes crucial

Power supply of the future

Storage and aggregated energy flexibility balance out intermittent renewable energy generation

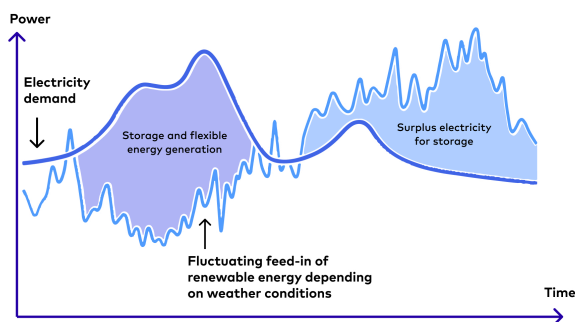


Illustration of how flexibility's service could affect future energy systems (Source: LichtBlick AG)

The concept of value stacking becomes paramount in the world of energy flexibility. While certain markets are still able to generate substantial revenues today by using an asset for a single explicit flexibility value stream, this will become less attractive over time. To counter this, accessing multiple implicit and explicit value streams with residential assets, which are currently only used for implicit flexibility will become crucial.

Here's a practical example of a household with a solar photovoltaic (PV) system and a battery storage unit: during periods of abundant solar generation, excess energy is stored in the battery to **optimize self-consumption**, limit the amount of power drawn from the grid and therefore minimize costs. At the same time, the battery can participate in grid balancing services by providing primary control power to the grid when frequency deviations occur. By unlocking both self-consumption optimization and grid services with the same asset, the owner can maximize their returns from the available flexibility without compromising on comfort.

The flexibility value chain



A schematic depiction of the energy flexibility value chain

The energy flexibility value chain involves a series of interconnected steps that need to be fulfilled to realize the full value of residential flexibility.

1. Access a customer base that owns assets and become prosumers.
2. Unlock the flexibility potential of assets through grid-edge technologies.
3. Provide technologies and services that are able to aggregate (and disaggregate) prosumer assets into a portfolio and stack use cases to access multiple value streams.
4. Forecast the available flexibility and optimization towards value streams – decide on the best asset for the value stream, and vice versa.
5. Access a marketplace in order to market and generate revenue from the available energy flexibility.

Roles in energy flexibility markets

Different actors are responsible for different parts of the energy flexibility value chain. Following the definitions given by [USEF](#), the most important roles to provide flexibility from residential resources are:

- **Balance Responsible Party (BRP):**
Ensures the balance between energy consumption and generation in a balancing group. Each physical connection point of the grid is associated with one BRP; in Germany, there are a few thousand BRPs.
- **Balancing Service Provider (BSP):**
Provides balancing services to help maintain the overall balance of the electricity grid; such as adjusting generation levels, activating reserve capacity, or participating in demand response programs.
- **Aggregator:**
Gathers and aggregates [flexibility from various sources](#), optimizing its use for market participation.

Activities assigned to the aggregator can be split into a technical and market actor.

- **Technical aggregator:**
Has the technical product and knowledge that allows for aggregation of resources and deployment of flexibility services.
- **Market aggregator:**
Can aggregate multiple technical aggregators to an asset portfolio and bid their resources as one large flexible resource on the market.

Accessible flexibility mechanisms for small-scale assets

Regulatory support and market structures that facilitate the integration of residential flexibilities into electricity markets are the key to success, but are not yet in place in all markets across Europe. The most promising use cases to leverage flexibility from residential [distributed energy resources \(DERs\)](#) are:

- **Grid balancing services** – Frequency containment reserve (FCR) & automatic Frequency Restoration Reserve (aFRR):
FCR is the fastest-acting grid balancing mechanism used to counteract sudden imbalances between electricity generation and consumption, helping to stabilize the grid frequency at 50 Hertz. aFRR is the secondary control reserve mechanism that acts automatically to restore the grid frequency to its nominal value after a disturbance. Residential flexibility can contribute to both FCR and aFRR by reacting quickly to imbalances in both demand and generation direction. Longer acting balancing services are mostly based on manual intervention and require larger controllable assets that can be dispatched by grid operators.
- **DSO congestion:**
DSOs can leverage residential flexibility assets to manage congestion in the local distribution

grid. By sending control signals to residential DERs, DSOs can alleviate stress on the local grid in times of high congestion. A notable regulation already in use since early 2024 to facilitate the DSOs job in this manner is [paragraph 14a of the Energy Industry Act \(EnWG\)](#) in Germany.

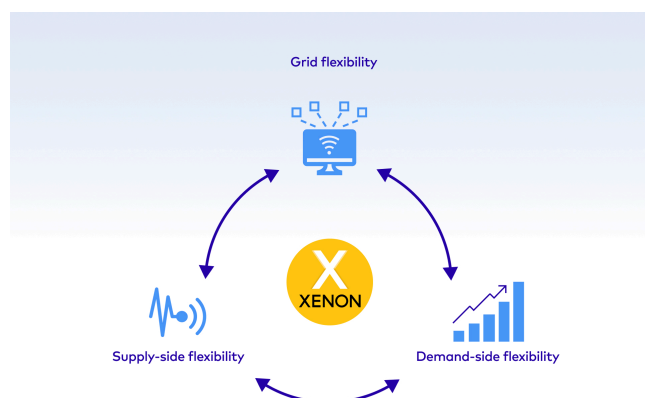
- **BRP portfolio imbalance optimization:**

BRPs need to manage discrepancies between forecasted and actual energy consumption or production within a portfolio of energy resources. Flexibility from residential DERs can help to minimize the imbalance in a BRP's portfolio by being used to counteract the observed imbalance. This not only assists BRPs in meeting their portfolio balancing requirements but also contributes to a more resilient and efficient energy system.

- **Day-ahead & intraday market optimization:**

Day-ahead and intraday market optimization for residential flexibilities refer to using residential flexibility to participate in electricity markets. In both day-ahead and intraday market optimization scenarios, effective communication and coordination between the DERs and market platforms are crucial. An [energy management system \(EMS\)](#) can enable residential users to actively participate in market-driven strategies.

How gridX unlocks flexibility



XENON plays a major role in unlocking energy flexibility

gridX unlocks the full potential of residential flexibility with a three-layered approach. Firstly, at the local level: optimization focuses on locally-produced electricity that requires a fast reaction time, such as fuse protection. Secondly, an EMS should also incorporate dynamic external signals, such as [time of use tariffs](#). These two layers represent implicit flexibilities, which are already operational. The third optimization layer will provide explicit flexibility services, such as balancing services, to different flexibility requesting parties like Transmission Service Operators.

Outlook

As of now, energy flexibility primarily originates from large facilities such as power plants and grid-scale [battery systems](#) because processes have been designed with larger assets in mind which makes it easier for them to participate in most flexibility markets. However, the landscape is evolving, and with a rising number of smaller-scale prosumers and decentralized energy assets, we at gridX expect a shift towards incorporating flexibility from more diverse sources.

gridX is dedicated to empower small-scale assets and offer a **digital infrastructure as foundation of future energy systems** that facilitates the seamless integration of residential energy assets into explicit flexibility services. Yet, a truly successful product in this space goes beyond mere functionality. To achieve this, **gridX seeks like-minded partners** who are committed to crafting a compelling solution for end-users – to create an innovative and user-friendly joint energy experience and thus makes a decisive contribution to the success of the energy transition.

Get the report!

Time of Use Tariff Report 2024 – This report looks at how time of use tariffs incentivize grid-friendly consumption to minimize electricity prices in multiple markets. Challenges remain in their adoption, but with a mindset shift, the right partnerships and advanced technology, the full potential of time of use can be tapped into.

[Read now](#)



gridX

Office Aachen
Dennewartstraße 25
52068 Aachen

Office Munich
Theresienhöhe 12
80339 Munich

Contact
info@gridX.ai
+49 (0) 241 412597 10



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E-mobility
OEMs
Installers
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